

PRACTICAL TECHNICAL ASSIGNMENT

Full Stack Developer Intern

DroneTV AI Support & Lead Assistant — Web Application

Prepared by	IPAGE Group — Talent & Engineering
Distribution	Sent to shortlisted candidates via Indeed
Assignment type	Take-home practical technical assessment
Maximum duration	3 days from receipt

This document outlines a self-contained take-home assignment used by IPAGE Group to evaluate hands-on full-stack engineering ability. It covers frontend, backend, database, security, and delivery expectations for candidates applying to the Full Stack Developer Internship programme.

1. Assignment Overview

IPAGE Group is evaluating candidates for the position of **Full Stack Developer Intern**. This practical assignment is designed to assess real-world, end-to-end web application development skills using a modern JavaScript/TypeScript stack, a backend API, and a persistent database.

Objective

Candidates must demonstrate the ability to design and build a real-world full-stack web application, evaluating proficiency in:

- React.js and TypeScript for building a typed, component-driven frontend
- Clean HTML & CSS with a fully responsive layout
- A backend API layer with proper REST conventions
- A relational or document database for persistent storage
- Git / GitHub for version control and submission

Reference Website (context only)

Candidates may review <https://dronetv.in> to understand DroneTV's services, training programmes, and general business context. The existing site must **not** be copied — candidates are expected to design their own original UI and implementation.

2. Main Assignment

Build a responsive **DroneTV AI Support & Lead Assistant** web application containing a chatbot that answers predefined questions about DroneTV and captures customer / student enquiries. A real AI/LLM API is **not required** — a predefined or rule-based response system is sufficient.

Part 1 — Frontend (React.js, TypeScript, HTML, CSS)

Candidates should build the following screens, fully responsive across desktop, tablet, and mobile:

- Landing / Home page
- Chatbot interface
- Services section
- Courses / Training section
- Contact / Enquiry form
- Simple Admin / Enquiry dashboard

Part 2 — Chatbot Functionality

The chatbot should support predefined questions such as:

What services does DroneTV provide?
What courses / training are available?
How can I contact DroneTV?
How can I register?
I am interested in a service.
I am a student.
I want to speak with someone.

The chatbot must also:

- Display both user and bot messages, maintaining conversation history for the session
- Provide suitable predefined responses to the questions above

- Handle unknown / unmatched questions with a graceful fallback response
- Include a clear / reset conversation option

3. Part 3 — Lead / Enquiry Collection

The chatbot must allow a user to submit an enquiry, collecting the fields below. Submissions must be validated before sending, with clear success and error messaging shown to the user.

Name	Email	Phone
User type (Student / Customer / Other)	Service or course of interest	Message

4. Part 4 — Backend API

Build the backend using **Node.js + Express.js** or **PHP**, exposing REST endpoints that support creating, reading, updating, and deleting enquiries using proper HTTP methods (POST, GET, PUT/PATCH, DELETE).

API Documentation

Method	Endpoint	Purpose
GET	/api/enquiries	Get all enquiries
GET	/api/enquiries/:id	Get a single enquiry
POST	/api/enquiries	Create a new enquiry
PUT / PATCH	/api/enquiries/:id	Update an enquiry
DELETE	/api/enquiries/:id	Delete an enquiry

Endpoint names may be adjusted to match the candidate's implementation.

5. Part 5 — Database

Use one database technology — **PostgreSQL**, **MySQL**, or **MongoDB** — with an appropriate structure for storing enquiries. Candidates should demonstrate a working database connection, full CRUD operations, input validation, and appropriate error handling.

6. Part 6 — Admin Dashboard

Build a simple dashboard where an administrator can:

- View and search enquiries
- Filter by Student / Customer
- View full enquiry details
- Change enquiry status
- Delete an enquiry

Possible enquiry statuses:

New	Contacted	In Progress	Closed
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7. Part 7 — Error Handling

- Empty required fields
- Invalid email format
- Invalid phone number
- Failed API request
- Server / database error
- Unknown chatbot question
- Non-existing enquiry ID

Sensitive technical error details must never be exposed directly to end users.

8. Part 8 — Security

- Validate input on both frontend and backend — never trust client-side validation alone
- Sanitize / escape all user-generated content
- Never expose database credentials
- Use environment variables for secrets
- Handle unauthorized or invalid requests appropriately

Destructive security testing is not permitted as part of this assignment.

9. Part 9 — GitHub & Submission

Upload the completed project to GitHub in a repository named **FullStack_Chatbot_Task_FirstName_LastName**. The README must include project description, features, technologies used, project structure, setup instructions, environment variables, database setup, API endpoints, screenshots, run instructions for frontend and backend, and a live demo link if available.

Submission Package (Google Drive Folder)

Create one Google Drive folder named **FullStack_Chatbot_Task_FirstName_LastName** containing the following subfolders:

01_Source_Code	02_Screenshots	03_API_Documentation	04_Database
05_Video_Walkthrough	06_GitHub	07_Resume	

The submission must include complete source code, database schema/instructions, API documentation, screenshots, the GitHub repository link, a 5–10 minute video walkthrough, and an updated resume in PDF format.

Video Walkthrough (5–10 minutes)

Demonstrate the frontend, chatbot, enquiry submission, API requests, database records, admin dashboard, search/filter, status updates, and error handling — and briefly explain the architecture: **Frontend → API → Backend → Database**.

Evaluation Criteria

React & TypeScript implementation, UI/UX and component structure, state management, API integration, backend & REST design, database design and CRUD operations, validation and error handling, security awareness, code organization, Git/GitHub usage, problem-solving, and overall application quality.

DEADLINE

Maximum 3 days from the date this assignment is received. AI coding tools may be used during development, but candidates must be able to explain any code they submit. Do not copy DroneTV's existing website.

For questions regarding this assignment, please reply directly to the Indeed message thread through which it was received. We look forward to reviewing your submission.